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LC T P Spring symposium: Theoretical Physics and AI

***Simplified* Reasoning Fin-Tuning: RRL & SFT**

- Lets try a lot and learn from my
success and failures

• This might take a while.....



- Lets see how a master would do it

- How likely could I have done that?



A movie poster for 'Edge of Tomorrow'. It features Tom Cruise and Emily Blunt in futuristic military armor, standing on a battlefield with a city in the background under a sky filled with flying machines and explosions. The title 'EDGE OF TOMORROW' is prominently displayed in the center, with 'EDGE OF' in white and 'TOMORROW' in red. Below the title is the tagline 'LIVE. DIE. REPEAT.' in white.

EDGE OF TOMORROW

LIVE. DIE. REPEAT.

RL



SFT

Simplified Reasoning Fine-Tuning: RL & SFT

RL

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- This might take a while....



SFT

- Lets see how a master would do it
- How likely could I have done that?



Reasoning Fine-Tuning: Outcome RL

Problem Prompt

Verifiable Solution

Response 1

Response 2

⋮

Response N

Generation